



Spring 2026

EMCH792 – Optimal State Estimation

Homework 3

Submission deadline: Thursday, 02/26/2026, 11:59pm EST

- Submission only through Blackboard, email submissions will not be accepted.
- Make sure you properly upload the file(s) and that you can successfully view your submission on BB. Keep a copy of the submission receipt you get from BB.
- Late submissions will be graded with the policy described in the syllabus.
- For questions email tsucin@email.sc.edu and copy vitzilaos@sc.edu

Homework Instructions

A. Written Portion

Solve the following textbook problems by hand and provide your solutions in a scanned PDF document.

1. Exercise 2.5, page 75 of the textbook

2.5 The probability density function of an exponentially distributed random variable is defined as follows.

$$f_X(x) = \begin{cases} ae^{-ax} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

where $a \geq 0$.

- a) Find the probability distribution function of an exponentially distributed random variable.
- b) Find the mean of an exponentially distributed random variable.
- c) Find the second moment of an exponentially distributed random variable.
- d) Find the variance of an exponentially distributed random variable.
- e) What is the probability that an exponentially distributed random variable takes on a value within one standard deviation of its mean?

2. Exercise 2.8, page 76 of the textbook

2.8 Consider two zero-mean uncorrelated random variables W and V with standard deviations σ_w and σ_v , respectively. What is the standard deviation of the random variable $X = W + V$?

3. Exercise 2.12, page 76 of the textbook

2.12 Suppose X is a random variable, and $Y(t) = X \cos t$ is a stochastic process.

- a) Find the expected value of $Y(t)$.
- b) Find $A[Y(t)]$, the time average of $Y(t)$.
- c) Under what condition is $\bar{y}(t) = A[Y(t)]$?

B. Computer Portion

In this homework you will learn to generate uncorrelated and correlated random variables in **MATLAB**. All the files that you will need for this homework are in the **ws** folder. You are expected to work inside this folder only. Open **MATLAB** and set the current folder to **ws**, and do the following:

- Start by opening the live script file “HW_3_manual.mlx” using **MATLAB**. This will guide you through basic **MATLAB** functions to analyze linear systems. Read through the file and understand how each function is implemented in **MATLAB**.
- Open the file “HW_3.m” and follow the instructions in the file to solve the problems below:

B1

2.15 Generate $N = 50$ independent random numbers, each uniformly distributed between 0 and 1. Plot a histogram of the random numbers using 10 bins. What is the sample mean and standard deviation of the numbers that you generated? What would you expect to see for the mean and standard deviation (i.e., what are the theoretical mean and standard deviation)? Repeat for $N = 500$ and $N = 5,000$ random numbers. What changes in the histogram do you see as N increases?

B2. Noise Covariance matrix calculation

Let $\dot{x}(t) = Ax(t) + B_u u(t) + B_w w(t)$ with $A = \begin{bmatrix} 3 & 1 \\ 0 & -1 \end{bmatrix}$, $B_u = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$, $B_w = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$.

Let $S_w = 0.11$, $\Delta_{t1} = 0.5$, $\Delta_{t2} = 0.01$.

For each Δ_t do the following (using MATLAB):

1. Use the Z block matrix to calculate the exact $\Sigma_{\tilde{w}}$ and the approximate $\Sigma_{\tilde{w}}$ and compare them. What do you notice in each case?
2. Using the Z block matrix, find A_d in each case. Verify your finding with the `c2d.m` function.
3. Compare predictions of steady-state performance: Using the `dlvap.m` and `lvap.m` functions, find $\Sigma_{\tilde{x},ss}$ in both continuous and discrete systems and compare them.

Submission Instructions

Your submission must include the following files:

- “HW_3.m” -> The edited MATLAB script with your solutions
- PDF document with the scanned solutions

Upload all files on Blackboard in a single submission.

Make sure that the scripts you submit can run. Don't copy the results that appear in the command window in your scripts and make sure that if you add any comments in your scripts that they follow the proper **MATLAB** syntax. **Scripts that do not run will not be graded.**

Grading Rubric

Problem	Points
A.1	20
A.2	20
A.3	20
B.1	20
B.2	20